

Mathematics Reference Sheet

1 Field Theory (Teorija polj)

1.1 Vector Operations

$$\text{Gradient: } \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \quad (1)$$

$$\text{Divergence: } \nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \quad (2)$$

$$\text{Curl (Rotor): } \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} \quad (3)$$

$$\text{Laplacian: } \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \quad (4)$$

1.2 Field Properties

$$\text{Irrotational: } \nabla \times \mathbf{F} = 0 \quad (5)$$

$$\text{Solenoidal: } \nabla \cdot \mathbf{F} = 0 \quad (6)$$

$$\text{Conservative: } \mathbf{F} = -\nabla \phi \quad (7)$$

$$\text{Potential: } \phi = - \int \mathbf{F} \cdot d\mathbf{r} \quad (8)$$

2 Spatial Geometry (Prostorska geometrija)

2.1 Coordinate Transformations

$$\text{Polar: } x = r \cos \theta, \quad y = r \sin \theta \quad (9)$$

$$\text{Cylindrical: } x = r \cos \theta, \quad y = r \sin \theta, \quad z = z \quad (10)$$

$$\text{Spherical: } x = r \sin \phi \cos \theta, \quad y = r \sin \phi \sin \theta, \quad z = r \cos \phi \quad (11)$$

2.2 Curves and Surfaces

$$\text{Arc length: } s = \int_a^b |\mathbf{r}'(t)| dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} dt \quad (12)$$

$$\text{Tangent vector: } \mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \quad (13)$$

$$\text{Normal plane: } \mathbf{r}'(t_0) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0 \quad (14)$$

2.3 Multiple Integrals

$$\text{Double integral: } \iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx \quad (15)$$

$$\text{Area: } A = \iint_D 1 dA \quad (16)$$

$$\text{Volume: } V = \iint_D f(x, y) dA \quad (17)$$

$$\text{Centroid: } \bar{x} = \frac{1}{A} \iint_D x dA, \quad \bar{y} = \frac{1}{A} \iint_D y dA \quad (18)$$

$$\text{Triple integral: } \iiint_E f(x, y, z) dV \quad (19)$$

$$\text{Mass: } m = \iiint_E \rho(x, y, z) dV \quad (20)$$

$$\text{Moment of inertia: } I = \iiint_E r^2 \rho dV \quad (21)$$

2.4 Line and Surface Integrals

$$\text{Line integral (scalar): } \int_C f ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| dt \quad (22)$$

$$\text{Line integral (vector): } \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \quad (23)$$

$$\text{Surface integral: } \iint_S f dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| du dv \quad (24)$$

2.5 Integral Theorems

$$\text{Green's: } \oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \quad (25)$$

$$\text{Stokes': } \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} \quad (26)$$

$$\text{Gauss (Divergence): } \iiint_V \mathbf{F} \cdot d\mathbf{S} = \iiint_E \nabla \cdot \mathbf{F} dV \quad (27)$$

3 Complex Analysis (Kompleksna analiza)

3.1 Complex Functions

$$z = x + iy, \quad |z| = \sqrt{x^2 + y^2}, \quad \arg z = \arctan\left(\frac{y}{x}\right) \quad (28)$$

$$e^z = e^x(\cos y + i \sin y) \quad (29)$$

$$\log z = \ln |z| + i \arg z + 2\pi i k \quad (30)$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2} \quad (31)$$

$$\sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2} \quad (32)$$

3.2 Analyticity

Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (33)$$

Harmonic condition: $\nabla^2 u = \nabla^2 v = 0$

3.3 Complex Integration

$$\text{Cauchy's formula: } f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz \quad (34)$$

$$\text{Residue theorem: } \oint_C f(z) dz = 2\pi i \sum \text{Res}(f, z_k) \quad (35)$$

$$\text{Laurent series: } f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n \quad (36)$$

4 Integration (Integrali)

4.1 Integration Techniques

$$\text{Substitution: } \int f(g(x))g'(x)dx = \int f(u)du, \quad u = g(x) \quad (37)$$

$$\text{By parts: } \int u dv = uv - \int v du \quad (38)$$

$$\text{Partial fractions: } \frac{P(x)}{Q(x)} = \sum \frac{A_i}{(x - a_i)^{n_i}} \quad (39)$$

4.2 Elementary Integrals

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 \quad (40)$$

$$\int \frac{1}{x} dx = \ln |x| + C \quad (41)$$

$$\int e^x dx = e^x + C \quad (42)$$

$$\int \sin x dx = -\cos x + C \quad (43)$$

$$\int \cos x dx = \sin x + C \quad (44)$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C \quad (45)$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C \quad (46)$$

5 Derivatives (Odvedi)

5.1 Differentiation Rules

$$(f + g)' = f' + g' \quad (47)$$

$$(fg)' = f'g + fg' \quad (48)$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad (49)$$

$$(f(g(x)))' = f'(g(x)) \cdot g'(x) \quad (50)$$

5.2 Elementary Derivatives

$$\frac{d}{dx} x^n = nx^{n-1} \quad (51)$$

$$\frac{d}{dx} e^x = e^x \quad (52)$$

$$\frac{d}{dx} \ln x = \frac{1}{x} \quad (53)$$

$$\frac{d}{dx} \sin x = \cos x \quad (54)$$

$$\frac{d}{dx} \cos x = -\sin x \quad (55)$$

$$\frac{d}{dx} \tan x = \sec^2 x \quad (56)$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2} \quad (57)$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}} \quad (58)$$